

ARTHROPODS

Jointed legs and flexible armour have helped a phylum that includes winged insects and underwater crustaceans to evolve unrivalled diversity.

There are more species of arthropods known to scientists than all the other animal phyla added together – and doubtless many more await discovery. They have an extraordinary range of lifestyles including grazing, predation, filtering particles of food in water, and drinking fluids such as nectar or blood.

Arthropods are coated by an exoskeleton made of a tough material called chitin. It is sufficiently flexible around the joints and between segments to allow for mobility, but resists stretching. As arthropods grow, they must periodically moult their exoskeleton, which is replaced by a slightly larger one. The exoskeleton serves as protective armour but also helps reduce water loss in very dry habitats.

BODY PARTS

Arthropods evolved from a segmented ancestor, perhaps a worm-like creature. Body segmentation persists in all arthropods, but is especially obvious

in millipedes and centipedes. In other groups, various segments have fused together into discrete body sections. Insects are divided into a sensory head, a muscular thorax with legs and wings, and an abdomen containing most of the internal organs. In arachnids and some crustaceans, the head and thorax are fused into one single section.

GETTING OXYGEN

Aquatic arthropods, such as crustaceans, breathe with gills. The bodies of most land arthropods – insects and myriapods – are permeated by a network of microscopic, air-filled tubes called tracheae. These open out through pores known as spiracles on the sides of the body, usually one pair per body section. Small muscles inside the spiracles regulate the flow of air by means of valves. In this way, oxygen can reach all cells of the body and does not need to be transported in blood. Some arachnids breathe with tracheae; others with book lungs in their abdomen that evolved from the gills of their aquatic ancestors. Most use a combination of the two.

PHYLUM	ARTHROPODA
CLASSES	19
ORDERS	123
FAMILIES	About 2,300
SPECIES	About 1.2 million

DEBATE CAMOUFLAGE AND MIMICRY

Many arthropods have adapted to their environment by blending imperceptibly into it. Stick insects, for example, look so like twigs that they are very difficult for predators to spot. In contrast, arthropods, such as wasps, have evolved bright colours that warn others they are distasteful or dangerous. The hornet moth is completely harmless but avoids predation because it looks and sounds exactly like a hornet. Despite their similar appearance, other anatomical features separate them into different orders. The hornet moth belongs to the order Lepidoptera and the hornet to the order Hymenoptera.

MILLIPEDES AND CENTIPEDES

Millipedes and centipedes form a group of multi-segmented arthropods called myriapods. Millipedes typically have two legs per body segment; centipedes have only one per segment. Whereas millipedes are vegetarian, centipedes are predatory carnivores.

2–3 mm
1/16–1/8 in



BRISTLY MILLIPEDE
Polyxenus lagurus
F: Polyxenidae

One of a family of millipedes armed with defensive bristles, this tiny species of the northern hemisphere lives under the bark of trees and in leaf litter.

WHITE-RIMMED PILL MILLIPEDE
Glomeris marginata
F: Glomeridae

Pill millipedes, such as this European species, have fewer body segments than other millipedes and can roll into a ball when disturbed.



AMERICAN SHORT-HEADED MILLIPEDE
Brachycybe sp.
F: Andrognathidae

This is a N. American representative of a family of small, flat millipedes that live in rotting wood and leaf litter.

segmented
exoskeleton
4–5 cm
1 1/2–2 in



BROWN GIANT PILL MILLIPEDE
Zephronia sp.
F: Zephroniidae

Giant pill millipedes, such as this Bornean species, differ from their smaller northern hemisphere relatives in that they have 13 body sections rather than 12.



3–4 cm
1 1/4–1 1/2 in

BLACK GIANT PILL MILLIPEDE
Zoosphaerium sp.
F: Sphaerotheriidae

One of many southern hemisphere giant pill millipedes, this species is native to Madagascar.

overlapping plates



frontal
shield

0.6–2 cm
1/4–3/4 in